BRAKE SYSTEM PLAUSIBILITY DEVICE - BSPD

LORENZO SERRANO LV APPLICATION

RESTRICTIONS:

- hardware only
- raw sensor signals straight to BSPD

BSPD MONITERS:

- hard braking AND throttle > 10% open FOR 7 1 3
- loss of braking sensor signal FOR 100 ms
- loss of throttle sensor signal FOR 100 ms

IF ABOVE EXIST:

- shutdown circuit

OP-AMP COMPARATOR

V1

open - loop configuration

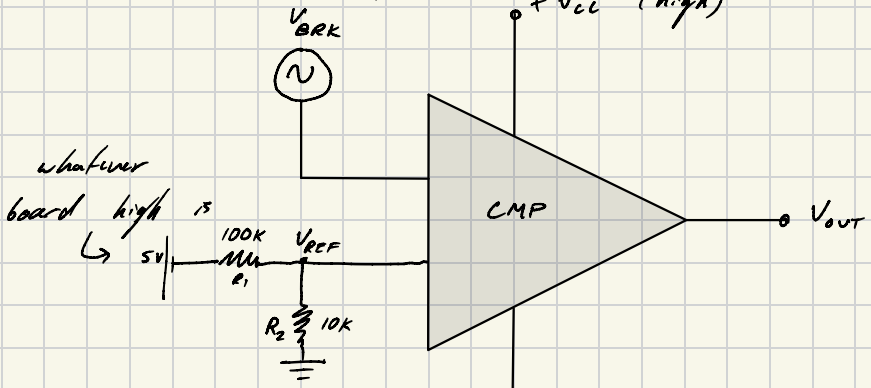
FUNCTION:

$V_{BRK} > V_{REF}$, $V_{OUT} = +V_{CC}$ (high)

$V_{BRK} < V_{REF}$, $V_{OUT} = 0V$ (low)

NOTE

V_{REF} divided input voltage used UCI's posted voltage (0.5V - 4.5V)



$V_{REF} = \frac{R_2}{R_1 + R_2} \times 5V$ pick $R_2 = 10K\Omega$ $V_{REF} = 0.45V$

$$.45 = \frac{5 \times 10{,}000}{R_1 + 10{,}000}$$

$$R_1 = \frac{50{,}000}{.45} - 10{,}000$$

$$= 101.1\ K\Omega$$

take $100K\Omega$ closest nominal.

so $V_{REF} = 0.455\ V$